SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV **COURSE CODE: DSA0202**

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts of image formation, pinhole to perspective camera models, homogeneous coordinates, and multi-view geometryand interpret real-world visual scenes using OpenCV or MATLAB. (BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20

Annexure A

Concept Mapping

Code of Conduct:

I P Sai Bhavyatha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

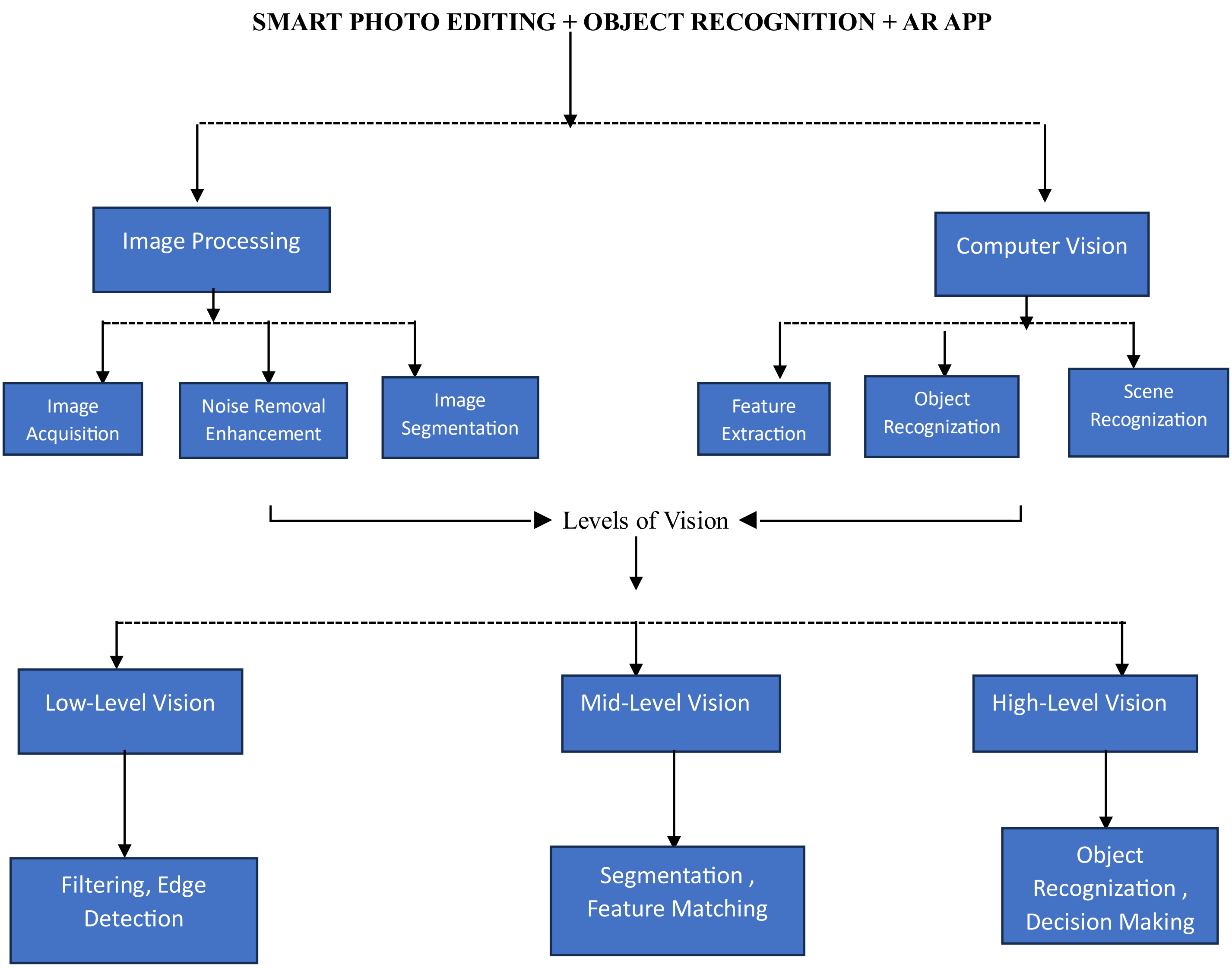
1. **A company is developing a smart photo editing + object recognition + AR app.**

Create a concept map connecting:

- **Image Processing**
- **Computer Vision**
- **Computer Graphics**
- **Levels of Vision**
- **Applications**

Show:

- **Workflow of the system**
- **Interconnections between components**
- **Real-world examples**



2. An autonomous drone captures images, calibrates cameras, estimates depth, and reconstructs a 3D scene. Draw a concept map linking:

- **Camera Model**
- **Image Formation**
- **Camera Calibration**
- **Binocular Vision**
- **Multiple View Geometry**

Include:

- **Flow of data**
- **Dependencies**
- **Output (3D reconstruction)**